

Data Handling: Import, Cleaning and Visualisation

Exercise 5: Data Preparation and Data Analysis with R

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Exercises¹

The GitHub repo for Exercise Session 5 contains csv files with data needed to solve the following exercises. Make sure to store them in your `r_course/data/` folder. Do not forget to load `tidyverse` before you get started!

Exercise A

Read `stocks.csv` into RStudio (store it in variable `stocks`). The dataset is currently in long format. Use `pivot_wider()` to transform it into wide format (separate columns for different years). Store the resulting tibble/data.frame in a variable called `stocks2`. Then, run the following code.

```
pivot_longer(stocks2, `2015`:`2016`, names_to = "year", values_to = "return") == stocks
```

Why is the result not identical to the original dataset (`stocks`)? In other words, why are `pivot_longer()` and `pivot_wider()` not perfectly symmetrical? (Hint: look at the variable types and think about column names.)

Exercise B

Read `countries.csv` into RStudio (store it in variable `countries`). Then, run the following code.

```
pivot_longer(countries, 1999, 2000, names_to = "year", values_to = "cases")
```

Why does this code fail? How can you fix the problem?

Exercise C

Read `people.csv` into RStudio (store it in variable `people`). Try to use `pivot_wider` on `people`. Why does it fail? How could you add a new column to fix the problem?

Exercise D

Read `preg.csv` into RStudio (store it in variable `preg`). Tidy `preg`. Do you need to use `pivot_wider` or `pivot_longer`? What are the variables?

¹Adopted from <https://r4ds.had.co.nz/tidy-data.html>.

Additional Exercises (advanced)

Exercise F

You want to investigate the relationship between GDP per capita and CO2 emissions per capita across countries. On Canvas, download the dataset `co2.csv`. This dataset has been downloaded from the website “Our World in Data” (<https://ourworldindata.org/grapher/co2-emissions-vs-gdp>). Investigate the relationship between the dependent variable `CO2_per capita` and the independent variable `GDP_per capita` using a linear regression of the following form:

$$\text{CO2_per capita}_i = \alpha + \beta \text{GDP_per capita}_i + u_i.$$

Estimate this linear regression by creating your own function `myols` that takes the following three arguments: (x) for the dependent variable, (y) for the independent variable, and *matrix*, which takes `TRUE` if the function estimates the coefficients using matrix algebra, and `FALSE` if the function estimates the coefficients using the sum operators. Compare the coefficients with the R built-in function `lm()`. Finally, comment and interpret the coefficients.

(Hint 1 (linear regression with sum operator): the slope parameter of the linear regression is expressed as the following formula:

$$\hat{\beta} = \frac{\sum x_i y_i - N \bar{y} \bar{x}}{\sum x_i^2 - N \bar{x}^2},$$

and the constant is given by this formula:

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x},$$

where $\bar{x}$ gives the mean of x and $\bar{y}$ gives the mean of y .

Hint 2 (linear regression with matrix algebra): the formula for the linear regression in matrix algebra is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}\mathbf{y},$$

where you find the inverse of a matrix with the command `'solve()'`, the cross-product of two matrices with `crossprod(X,Y) = t(X)%*%Y'`. Note that you need a column of 1s in your matrix of covariates to compute the intercept.)

(Note: this exercise is about programming and implementing the formula of the linear regression. You don't need to understand the mechanics of linear regression for the exam. This will be covered next semester in the course `Data Analytics II`.)

Mock Exam Questions